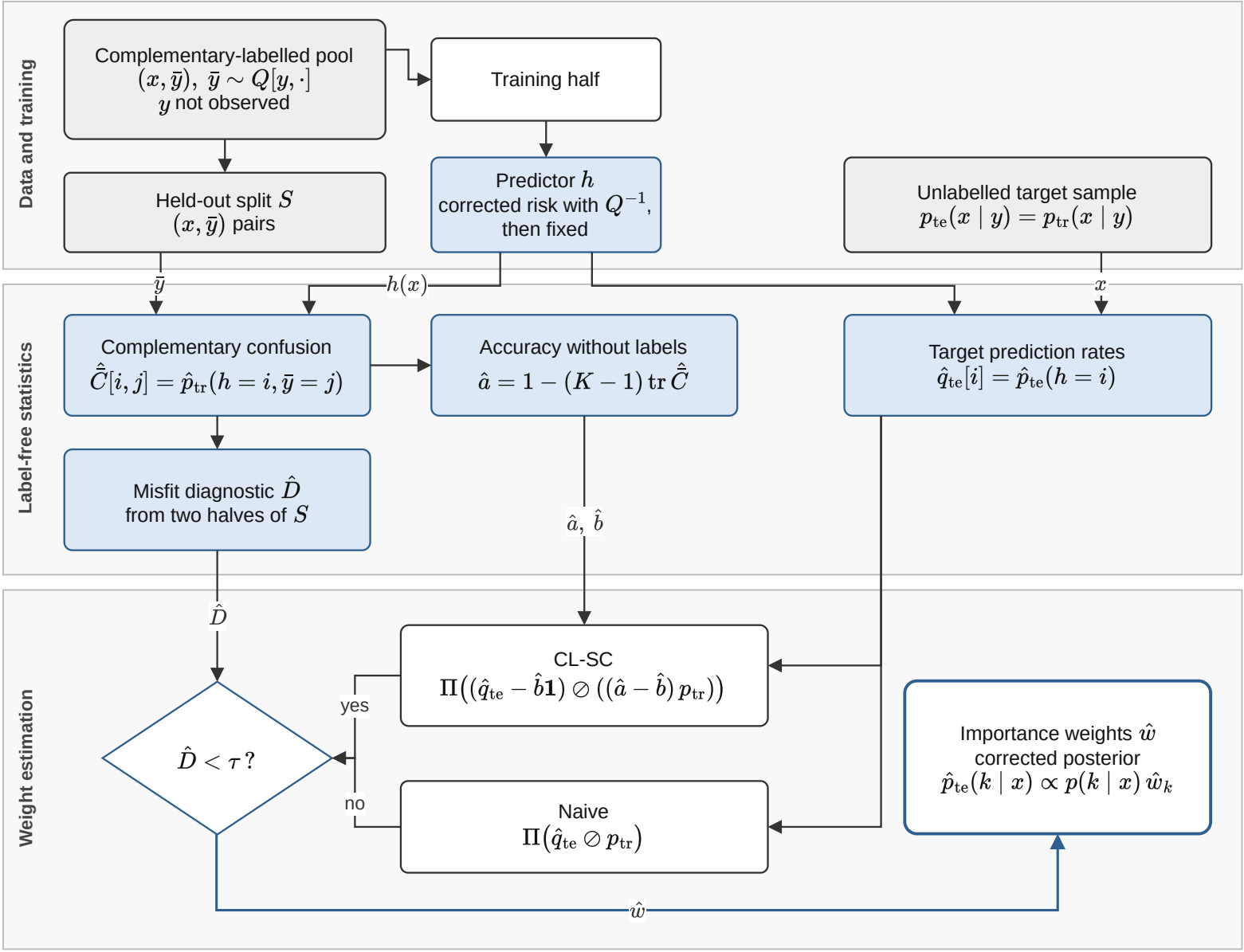

Data and training
Complementary-labelled pool
$(x, \bar{y})$, $\bar{y} \sim Q[y, \cdot]$
y not observed
Training half
Held-out split S
$(x, \bar{y})$ pairs
Predictor h
corrected risk with Q^{-1},
then fixed
Unlabelled target sample
$p_{\mathrm{te}}(x \mid y) = p_{\mathrm{tr}}(x \mid y)$
$\bar{y}$
$h(x)$
x
Label-free statistics
Complementary confusion
$\hat{\tilde{C}}[i, j] = \hat{p}_{\mathrm{tr}}(h = i, \bar{y} = j)$
Accuracy without labels
$\hat{a} = 1 - (K - 1)\,\mathrm{tr}\,\hat{\tilde{C}}$
Target prediction rates
$\hat{q}_{\mathrm{te}}[i] = \hat{p}_{\mathrm{te}}(h = i)$
Misfit diagnostic $\hat{D}$
from two halves of S
$\hat{a}$, $\hat{b}$
Weight estimation
$\hat{D}$
CL-SC
$\Pi\big((\hat{q}_{\mathrm{te}} - \hat{b}\mathbf{1}) \oslash ((\hat{a} - \hat{b})\, p_{\mathrm{tr}})\big)$
yes
$\hat{D} < \tau$?
no
Naive
$\Pi\big(\hat{q}_{\mathrm{te}} \oslash p_{\mathrm{tr}}\big)$
Importance weights $\hat{w}$
corrected posterior
$\hat{p}_{\mathrm{te}}(k \mid x) \propto p(k \mid x)\, \hat{w}_k$
$\hat{w}$